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Data mining techniques for analyzing bank customers: A survey

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Abstract. In today's business world, identifying the customers and analysis of their behavior is important for banking industry. Customer Relationship Management (CRM) is the process of maintaining profitable customer relationships by delivering customer value and loyalty. Moreover, CRM helps to improve the business relationships with customers. The goal of CRM is to maximize the lifetime value of a customer to an organization. Customer Lifetime Value (CLV) can rank and classify customers based on their lifetime value to identify valuable customers and retain them. There are several models for CLV estimation using the past data of customers. This subject helps organizations in their attempts to retain valuable customers. The banks must use appropriate data mining techniques to extract pattern and information from the existing data to gain competitive advantage. Therefore, data mining techniques have an important role to extract the hidden knowledge and information. The goal of this study is to review data mining techniques used for analyzing bank customers in order to help the banks to better identify their customers and design more efficient marketing strategies. The literature covered in this paper is related to the past seventeen years (2001–2017) and these approaches are compared in terms of data sets, prediction accuracy, and so on. We also provide a list of data sets available for the scientific community to conduct research in this field. Finally, open issues and future works in each of these items are presented.

Keywords: Customer relationship management, customer lifetime value, bank industry, data mining, loyalty

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1. Introduction

In today's business environment, it is essential to identify and analyze the customers' needs to gain a competitive advantage. Managers should try to retain customers and focus on key customers in order to increase their costs and gain profit. Today, customers share their negative experiences together using communication technologies, which can lead to the loss of current customers' trust. Organizations can identify customers and meet their needs to increase customer loyalty. Therefore, retaining key customers for banks is more beneficial than attracting new customers [1]. In today's continuously changing competitive business environment, the organizations have to analyze and understand customer needs and behavior. "Customer dynamics" is one of the most important issues to be considered when analyzing customer behavior, because customer behavior is often complex and uncertain in today's dynamic situation. Considering the dynamics of large organizations can cause improvements [2] and organizations with an understanding of customers' behavior can improve their marketing strategy using behavioral scoring models that help to analyze the behavior of customers [3]. Obtaining customer satisfaction is a modern approach for quality control in organizations. To reinforce customer orientation, many organizations choose customer satisfaction as their main performance indicator, although it is almost impossible to achieve [6]. The loss of customers in an enterprise can reduce its profits. The cost of attracting a new customer is five times higher than that of retaining the existing subscribers. Therefore, customer retention is a core Customer Relationship Management (CRM) issue. CRM creates a strong link between the organization and customers, which finally increases customer loyalty, and enhances return of investment. Organizations can boost customer relationship management and prevent the loss of key customers [4,57], an improve-

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ment that leads to competitive advantages. Companies recognize that CRM is a fundamental tool for building customer value and this topic helps to increase enterprise value [5,54]. Organizations try to increase Customer Lifetime Value (CLV) through CRM. There are many methods for evaluating the CLV that can be used by organizations to gain profit. The spectrum of research goals is shown in Fig. 1.

In Fig. 2, a timeline of researches in the literature is shown. For this propose, this paper reviews the researches belonging to CRM and CLV. The rest of the paper is organized as follows: in Section 2, CRM, CLV, and customer segmentation are briefly introduced. In Section 3, the data sets used in recent researches are reviewed. Section 4.1 introduces data mining (DM) techniques applied on CRM, CLV, and customer segmentation and in Section 4.2, the literature is categorized according to the techniques. We will analyze these researches from different aspects and try to provide a comprehensive overview of the literature on this subject, which can be helpful for researchers interested in this field. Evaluation results of the studied papers are illustrated in Section 4.3. In Section 5, we will pay attention to open issues in this context. Finally, we will comprehensively discuss the researches and their results in Appendix II. This information is collected in one table and can thus be easily compared.

2. Literature review

2.1. CRM and e-CRM

Customer Relationship Management (CRM) is a strategy that allows a business to manage its customer relationships. CRM is known as a process for improving the relationship between organizations and customers. It is a practice for creating tolerable connections between an organization and its customers. The goal of CRM is to forge closer and deeper relationships with customers. The most important factors that affect customer loyalty are satisfaction, trust, commitment, perceived value, perceived quality, intuitive image, empathy and switching barriers. Through management of relationships, CRM attempts to attract and retain customers, increase customer loyalty, customer satisfaction, and creation value for customers. According to studies, the cost of attracting a new customer is about five times more than retaining an existing one. Therefore, enterprises found that it is critical to develop a long-term relationship between themselves and their customers to achieve profitability and customer satisfaction in the long run [7,10,37,38,50–52]. CRM includes four dimensions: customer identification, customer attraction, customer retention, and customer development. For example, customer satisfaction has many benefits for firms, such as enhancing firm reputation, attracting and retaining customers, increasing customer loyalty, etc. [8,13,15]. Note that in [9], the authors surveyed the existing techniques in the field of CRM and explored some of the main challenges. However, this review did not focus on banking industry and had a general scope. In fact, performance evaluation and CRM in the banking industry are more complicated and involve several factors [49].

The use of Internet for commerce presents an opportunity for businesses to use it as a platform for the delivery of CRM functions on the Web (e-CRM). E-CRM expands the traditional CRM techniques by integrating new electronic channels such as web, wireless, and voice technologies, combining them with e-business applications into an overall enterprise CRM strategy. E-CRM is the adaptation of CRM in an e-commerce environment and helps to create and maintain customer relationship using the net [11,12].

2.2. CLV

Customer Lifetime Value (CLV) is a pattern of analytical CRM. The goal of CRM is to maximize the lifetime value of a customer for an organization. Evaluation of CLV is important for enterprises in terms of marketing to have an effective CRM. Studies have shown that the past behavior of a customer may not always predict their future behavior. Therefore, we need a metric to predict the future profitability of a customer. CLV is a metric that represents the total net profit of a company from any customer. CLV is a concept in CRM domain for evaluating the customer's value. In addition, it is an important metric for determining how much money a company intends to spend to acquire new customers and how much repeated business a company can expect from certain consumers. CLV can help you understand your most valuable user segments based on customer value over time and is thus considered as an important subject. CLV can rank and classify customers based on their lifetime value for identifying valuable customers and retaining them, as well as identifying and comparing market segments [7,13,17,18,58,59]. There are several models to estimate CLV using the past data of customers through which we can determine the customers

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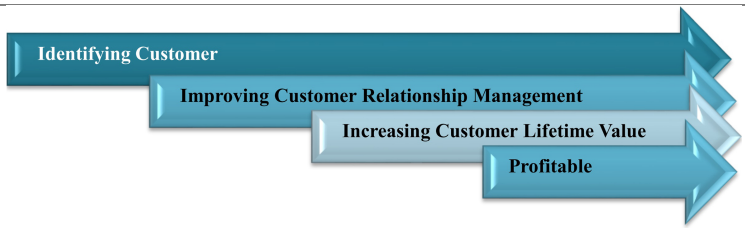


Fig. 1. Research' goals spectrum.

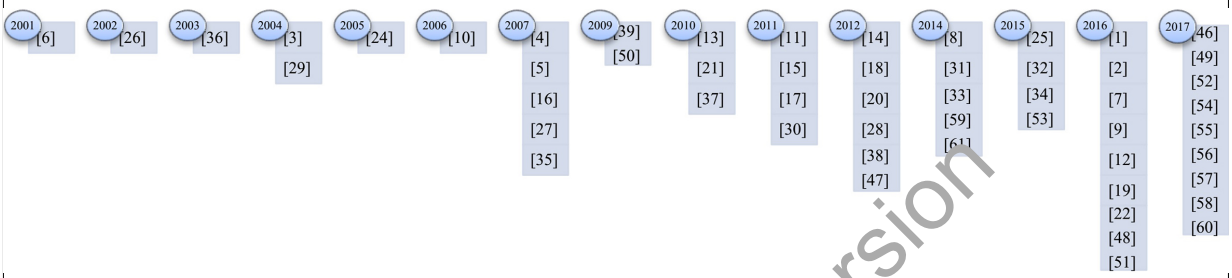


Fig. 2. Timeline of the researches in the literature.

who are more profitable than others. This topic helps the firms to retain their valuable customers [16]. CLV models are used to evaluate customer loyalty, recommend right products to right customers, and provide individual marketing decisions for each customer. Some CLV models are based on past behavioral models and some on future customer revenue. In addition, a number of CLV models can prospect the future monetary value of customers [14].

In the following, a summary of the some key CLV models are provided, which are divided into tree corresponding categories [17,18]:

1. Models for calculation of CLV: This category includes models that are specifically formulated to calculate the CLV.
2. Models of customer base analysis: The analysis of past behavior of customers to predict their future behavior.
3. Normative models of CLV: These models have been developed to understand the issues concerning CLV.

2.3. Customer segmentation

Customer segmentation is the process of dividing customers into distinct and meaningful groups of homogeneous customers on the basis of common attributes. Customer segmentation increases customer satisfaction and the company's expected profit. Customer segmentation is so important for companies to develop effective marketing strategies. The use of various marketing strategies leads to an enhanced value of customers.

Customer segmentation can be done based on various criteria, including degree of customer loyalty, purchase frequency, purchase volume, demographics, etc. The banks can differentiate themselves from their competitors through customer segmentation. Nowadays, customer segmentation has become important in the banking industry, and much research has been done in this respect. On the one hand, customers are demanding quality services from their banks, so that the banks have to keep their customers by customer segmentation to enhance the quality of their services. On the other hand, identifying valuable customers is an important issue and segmentation is an appropriate method to recognize the customers. Cluster analysis is defined as "partitioning data into meaningful subgroups, when the number of subgroups and other information about their composition may be unknown". There are many models for customer segmentation like SOM (Self-Organizing Map), fuzzy method, RFM (Recency, Frequency and Monetary), combination of some models (RFM, AHP, and K-means algorithm) and so forth [2,19–22,59,61].

3. Data sets

All the researches about CRM and banking industry have used a data set. They applied several DM tech-

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Table 1
Research' goals spectrum

Reference	Number of customers/ transactions	Number of clusters	Metrics	Data source
[1]	120	2	– Length of the relationship – Recency – Frequency – Monetary	The data from 120 customers were collected and normalized.
[22]	12,429	5	– CustomerID, – SellerID, – Request date, – ComID (commodities ID) – Box fields	The data include 500,220 sales records in sale of 179 different commodities to 12,429 customers.
[19]	250	5	– Gender – Age – Monthly income – Account type – Account closure history – Account currency – Location	The data set contains information about different characteristics of 250 loan applicants of a commercial bank located in Isfahan Province of Iran. The data were collected during a one-month period.
[7]	3,091	9	– Recency – Frequency – Monetary	The data, including date of last purchase, the frequency and money spent in purchases, are extracted from the customers' transaction database, from which 3,091 records remained after excluding incomplete data.
[2]	Step 1: 165,800 Step 2: 20000	3	Step1: – Recency – Frequency – Monetary Step2: – T – Recency – Frequency – Monetary	The data source has been the database of company.
[29]	1000	4	Classification ∗ good credit ∗ bad credit – Segmentation ∗ credit amount ∗ credit history ∗ duration in month ∗ other debtors and guarantors ∗ other installment plans ∗ present employment ∗ present residence ∗ property, purpose ∗ savings accounts and bonds ∗ status of existing checking account	
[27]	6.2 million	22	– Age – Demographics/ lifestyle data – Product ownership (type and intensity) – Activity level	Data set is based on the analysis of homogeneous groups instead of individual customers.

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Table 1, continued

Reference	Number of customers/ transactions	Number of clusters	Metrics	Data source
[10]	16,384	No	– Customer values – Potential value – Customer loyalty	The data of this study consist of 6-month service data of a wireless communication company in Korea. This data set is composed of 200 data fields and 16,384 records of customers.
[61]	10000	No	– Customer satisfaction – Customer loyalty – Product usage	The data set consist of the monthly data of 10,000 customers of a bank.
[11]	120	No	– Large bank – medium banks	The data from 120 customers of Thai commercial banks.
[35]	578	No	– Satisfaction – Likelihood of switching	The data include 578 existing customers of five major banks in a US state on the east coast.
[53]	45211	17	– Accuracy rate – Sensitivity analysis – Specificity	The data include 17 campaigns of a Portuguese bank conducted between May 2008 and November 2010.
[5]	149 articles	4	– Marketing – Customer service – Understanding customers	The data from four retail trade publications were analyzed over a 5 year period (2000–2005).
[25]	45,211	2	– Classification accuracy – Sensitivity – Specificity	The data include 17 campaigns of a Portuguese bank conducted between May 2008 and November 2010.
[36]	274	No	– Size of banks	The data come from 278 small and large retail banks in Denmark, Finland, Norway and Sweden.
[6]	160	No	– Personnel of the bank – Products – Image of the bank – Service – Access	Input data consist of 160 private customers and 95 companies.
[15]	7,000	4	– Recency – Frequency – Monetary	Data on purchase transactions, approximately 7,000 records, have been collected.
[17]	50,000	[illegible]	– Recency – Frequency – Monetary	The data were row data of 50,000 customers' transactions from early spring 2008 to late summer 2009.
[14]	22086	3	– Recency – Frequency – Monetary	Data set contains 65,535 transaction records with 22,086 customers.
[31]	296	5	– Repayment behavior – Transaction – Frequency – Monetary – Recency	296 customers of the bank in 2009 were used as the data set.
[21]	55,211	No	– Education – Days –segments – Times-Segments – Age range – Sex – Amount of transaction	A table containing effective debit card account information of 55,211 customers until October 2009, and another table storing over 11.3 million individual transaction records for these accounts from March 2007 to October 2009.

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Table 1, continued

Reference	Number of customers/ transactions	Number of clusters	Metrics	Data source
			– Transaction type – Terminal type – Occupation – Block code	
[16]	214	8	– Recency – Frequency – Monetary	Data set contains 214 RFM data from customers.
[20]	452	7	– Recency – Frequency – Monetary	Data related to 18-month transactions of the short term accounts of 452 customers whose deposit was about 50 million Rails.
[28]	10,281	4	– Recency – Frequency – Monetary	Data collected during 1997 to 1998 in Singapore, to measure customer's loyalty during these two years.
[18]	298	No	– Relation length – Recency – Frequency – Monetary	The source data is the real transaction data in the bank, which has 298 observed values collected in the file.
[34]	1000	No	– Recency – Frequency – Monetary	The data for this study include a selection of 1000 best customers.
[17]	50,000	9	– Recency – Frequency – Monetary	Two years' data of transactions, approximately five million records for 50,000 customers, have been collected with simple stochastic sampling method.
[56]	10471	9	– Length – Recency – Frequency – Periodicity	The data contain almost 2 million purchase transactions of 16024 customers during the period between October 1, 2012 and August 31, 2014.
[33]	– SCPOS-all data set 263,082 – Foodmart data set 251,395	No	– Recency – Frequency – Monetary	The final SCPOS-all data set contained 263,082 transactions and 12,505 items while the final Foodmart data set contained 251,395 transactions and 1560 items.

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niques to achieve their goals. In this study, data has been collected from banks and some data are taken from other industries. In Table 1, all of these data sets are shown in brief.

4. Data mining and its techniques on CRM, CLV, and customer segmentation

4.1. Data mining (DM) techniques

DM techniques can help a company to discover and extract useful information from their data warehouses, which cannot be discovered directly. This information can be used for the prediction of their customer behaviors. DM plays an important role for improving customer relationship management. Organizations use DM to improve their competitive advantage and add value to the customer. Therefore, organizations can discover customer needs to increase competitive advantages, respond to the expectations of customers and offer quality service. In recent decades, DM has attracted a lot of attention in data science for diagnosing heart diseases [23,52], designing software architecture [40–43], selecting design pattern [44,62], traffic accident prediction [63], fraud detection [64], ATM

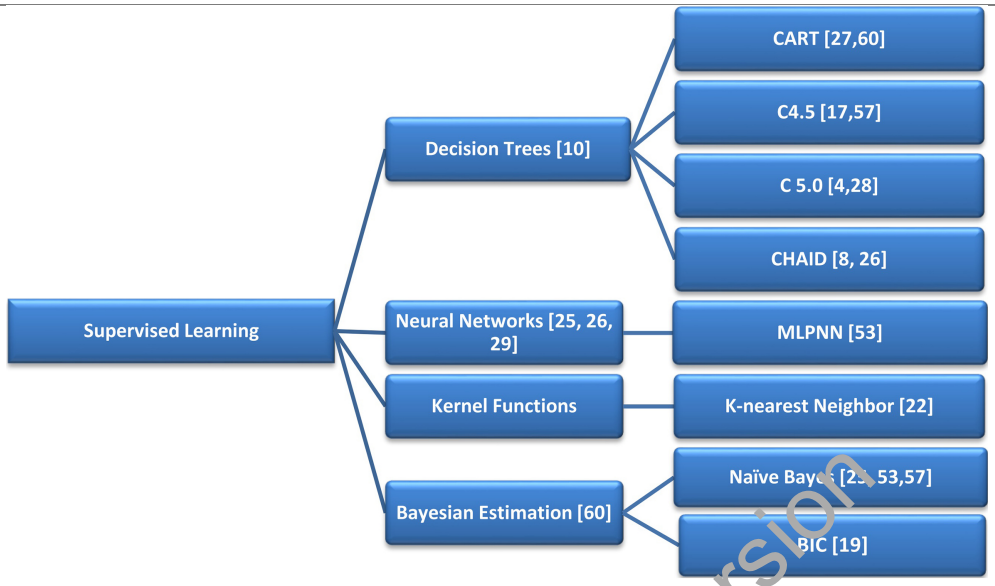


Fig. 3. Supervised techniques used in CRM, CLV and bank customer segmentation.

(Automated Teller Machine) management [65] and so on, and its techniques have been applied on almost every subject to analyze data and get accurate and reliable results based on the defined goals. Many studies have employed DM to analyze customer data but some have attempted to discover new DM techniques [5,8,17]. The important issue is how we can gain better insight and knowledge through using DM techniques to improve CRM and increase CLV and thereby create more profit for customers and company [2]. DM is a technology for extracting information from customer data, which is useful to identify customer demand, effectively promote customer value, and predict their future behavior. DM can also be considered as a process and a technology that uses statistical algorithms to discover hidden patterns and information from the existing data to gain competitive advantage [9,25,26]. Organizations have started using DM technologies to gain customer loyalty and increase the contribution of customer value [46].

DM techniques have been used to predict customer behavior and can obtain previously unknown and potentially useful information (including knowledge rules and regularities) by searching through a database. DM is a stage in Knowledge Discovery in Databases (KDD). In today's competitive business environment, organizations can improve their competitive advantage by using DM. Various applications are used for marketing, finance, and banking. Applications in these domains help to collect and storage a large amount of data. DM gains models from database and discovers patterns and correlations in data and can discover useful customer behavior patterns from large data repositories. In a dynamic business environment, the companies have to analyze and understand customer needs and behavior. Organizations need to have a deeper understanding of customer behavior. Today, customer behavior is often uncertain and changes over time, so organizations can use a number of methods to predict changes in customer behavior and the company needs to focus on mining changes in databases [2,15,24,26]. As we know, retaining a current customer and acquiring a new one is an important matter in banking industry. Therefore, we should use a number of methods for analyzing customer behavior, which are based on bank databases. Analyzing a bank customer is a difficult task because the bank databases are multi-dimensional and are comprised of monthly account records and daily transaction records. In addition, there are a variety of methods to analyze the customer behavior. In this regard, selection of the appropriate method for analysis is important.

In this paper, we review some DM techniques for analysis of bank customer behavior. DM techniques are divided into several groups as follows:

1. Supervised
2. Unsupervised
3. Evolutionary learning
4. Other

In the following, we discuss these techniques.

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4.2. Data mining techniques used in the literature

4.2.1. Supervised

Several supervised techniques are used in knowledge extraction and some of these techniques have been applied on the bank industry, which are shown in Fig. 3.

Decision tree is a supervised classification technique in DM. Decision tree is a predictive model that can be regarded as a tree. This technique represents sets of decisions. The decision tree is most often used for classification of a data set and can produce understandable rules. Specific decision tree techniques such as Classification and Regression Trees (CART), Chi Square Automatic Interaction Detection (CHAID), and Commercial version (C5.0) are highly efficient [10]. In [8], CHAID and C5.0 decision tree techniques are used for classification of customers and help the banking industry to make decisions. They provided a set of rules that can be applied to a new data set to predict which records will have a given outcome. CHAID analysis is an algorithm used for discovering the relationships between variables. CHAID analysis builds a model or tree to help determine how variables best merge to explain the outcome for a given dependent variable. In practice, CHAID is often used in direct marketing to understand how different groups of customers might respond to a campaign based on their characteristics. In [26], authors used CHAID through Applied Matrix for an initial segmentation modeling. Applied Matrix helps companies increase their competitive advantage, the lifetime value of customer, and decrease cost of customer acquisition. In this study, CHAID identified market segments that were formed by interactions among predictors of a chosen criterion variable like customer age. Another technique of decision tree is C5.0, which is a later version of C4.5 and can use continuous data, information theory, and learning method to build a decision tree. C5.0 creates decision tree from a data set by using Gain and Gain Ratio parameters. In [28], authors have used the C5.0 model to produce rules for predicting the level of loyalty based on demographic variables, on the obtained clusters from k-means and two-step algorithms. Moreover, in [4], C5.0 has been used as a classification technique. CART is a classification tree and a form of binary recursive partitioning. In [27], CART analysis has been used to build a regression tree, which uses the customer base into a set of homogeneous sub-groups for clustering. This technique was applied to determine membership of a set of classes as a function of certain predictor variables.

Neural network is an information-processing technique to capture and represent complex data relationships. A neural network acquires knowledge through learning. Neural networks are organized as a layer set of neurons, which include input layers, hidden layers, and output layers, and the computations of the network are performed in the hidden layer. The layers are made up of a number of interconnected nodes. Neural network has been used for creating predictive models such as customer lifetime value. Neural network has a wide range and can be applied to both supervised and unsupervised DM and to solve estimation problems [25,26]. In [29], neural networks have been used as the classification model and created a credit scoring. The neural networks are built with the data of the existing customers. Then, all these existing customers are evaluated by the model in order to detect their predicted credit status, which can be good or bad.

In [22], K-Nearest Neighbor (KNN) technique is used to classify and identify the goods that are more favorable to customers, which uses a database in which the data points are separated into several classes to predict the classification of a new sample point. In [22], the recommendation systems designed based on a combination of CF (Clustering features) and KNN were used in business.

In [25], Naïve Bayes (NB) classifier is used, which is based on Bayes' rule as a simple probabilistic classifier. The researcher used Bayes' rule as the basis for designing learning algorithms. Bayesian classifiers predict the probability that a sample belongs to a particular class. This technique has high accuracy and fastness to train with simple models and is thus used for large databases. According to [19], if the number of clusters is not determined, we can calculate their number by using Bayesian Information Criterion (BIC) and Akaike Information Criterion (AIC).

4.2.2. Unsupervised

Unsupervised is a DM technique for extracting hidden structure from unlabeled data, which is shown in Fig. 4. It does not require the humans to have a foreknowledge of the classes, and mainly uses clustering algorithms to classify an image data.

In [2], the GSP (Generalized Sequential Pattern) algorithm is used to extract the sequential patterns. It is a popular algorithm used for sequence mining, which extracts the patterns that appear more frequently than a user-specified minimum support while maintaining their item occurrence order. In [21], Apriori algorithm is used to determine the group of customer by creat-

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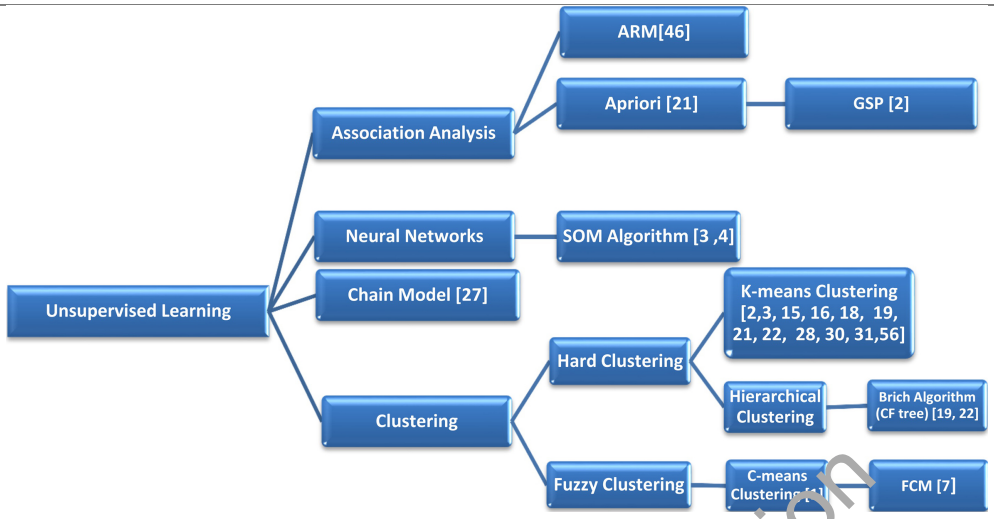


Fig. 4. Unsupervised techniques used in CRM, CLV and bank customer segmentation.

ing customer profiles and finding the relevant clustering rules.

In [3], SOM (Self-Organizing Map) has been used. It is an unsupervised learning technique that relates multi-dimensional data. This technique is an artificial neural network algorithm that contributes to mapping high-dimensional data into a two-dimensionally represented space. SOM has been used in a wide range of applications, including financial data analysis, medical data analysis, time series prediction, and industrial control. It is built with customer's data, which include variables from account and transaction data sets. In this model, all the customer's data are used to build the behavioral scoring model in order to predict potential customer behavior. In SOM, it is difficult to predefine a network size without obtaining prior knowledge about the organization of the data to achieve acceptable results. Therefore, the use of a Growing Hierarchical Self-Organizing Map (GHSOM) has been suggested.

Chain model is another unsupervised technique. In [27], a model has been used with a combination of first-order Markov chain modeling and CART. This model is based on the analysis of homogeneous groups instead of individual customers. The chain model has been used in marketing, including customer lifetime valuation.

Clustering is a popular unsupervised learning technique. It is used for finding classes or groups of a data set with most similarities in the same cluster, while the dissimilar objects are in a different cluster. Clustering is a DM technique used to divide data into related groups without a foreknowledge of the group definitions. Clustering techniques are classified as hard clustering and fuzzy clustering. In fuzzy cluster or soft cluster, each point may belong to two or more clusters with different degrees of membership. The well-known hard clustering algorithm (K-means) and Fuzzy clustering algorithm are mostly based on Euclidean distance measure. FCM algorithm is a well-known Fuzzy Clustering method that permits one point to belong to two or more clusters based on fuzzy logic. In [7], FCM algorithm is used to cluster data into nine optimum clusters based on three values of recency, frequency, and monetary. In [1], fuzzy clustering was applied to collect and normalize data from 120 customers based on four different variables, namely length of the relationship, recency of trade, frequency of trade, and monetary value. K-means is an unsupervised DM technique and the most popular hard clustering technique that divides data into groups, and the objects in each cluster are very homogenous and dissimilar with other clusters. Each data point belongs to only one cluster. This technique requires previous knowledge about the number of clusters. It takes the input parameter k (the number of clusters) and partitions a set of n objects into k clusters. In [2], K-means algorithm has been used and k objects have been randomly selected as the initial centers of clusters. Other objects were assigned to the closest clusters based on the distance between the object and the center of cluster. In [15], K-means algorithm was applied for customer segmentation in order to assess CLV for each segment. In [30], K-means algorithm was used as clustering algorithm for clustering the bank's customers. K-means algorithm was executed eight times and Dunn index was calculated in each time, because this algo-

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Fig. 5. Evolutionary techniques used in CRM, CLV and bank customer segmentation.

rithm needed this calculation. The optimum number of clusters (k) was found to be four. K-means has been often used as a credit and behavioral ranking model for customers of banks. These models classified customers into different clusters and K-means was applied for ranking new customers in terms of credits using a number of determinants like age, marital status, income, etc. The banks used this ranking for predicting future purchasing behavior of customers or their credit status [31]. In [31], K-means was used to build clusters of customers through RFM variables. To compute K-means, first all m objects were portioned into K initial clusters. Then, the objects whose Euclidean distances were nearest to the mean of cluster were assigned to that cluster. Centroid was recalculated for the clusters with a new item or a lost item. The process was repeated until there would be no more reassignment. In [21], K-means algorithm was used to divide customers based on recency, frequency, and monetary behavioral scoring predictors into three major profitable groups of customers: pamper user, transactor user, and raring user. As already mentioned, calculating bank customer's CLV is an important matter. Therefore, in [16], the authors used K-means clustering to determine CLV and segment customer based on recency, frequency, and monetary. Moreover, K-means algorithm was applied for cluster analysis of bank data to group customers in [18]. First, raw data were separated into seventeen clusters as favorites; then, the distance of each customer to the center of its cluster and the error function were calculated.

CF tree is a height-balanced tree that stores the clustering features for a hierarchical clustering. Each entry in CF tree represents a cluster of objects. In [19,22], the authors have used CF tree. For example, in [22], CF is a well-known technique in recommendation systems, which can be categorized as neighborhood-based and model-based techniques. The CF is calculated based on the nearest distinctive neighbor for each cluster of customers.

4.2.3. Evolutionary learning

As shown in Fig. 5, another DM technique is evolutionary learning. This technique can be used in any classification-based prediction scenario. In addition, it helps to predict the value of a user-specified goal attribute based on the values of other attributes. For example, the banks have used this technique to predict credit. Genetic algorithm (GA) is a meta-heuristic algorithm used for data clustering. In [1], this algorithm was used for customer clustering.

4.2.4. Other

In addition to the above-mentioned techniques, some researches have attempted to suggest alternative techniques, which we have categorized as "Other" and are shown in Fig. 6.

Delphi method is used in [32] for finding actual CRM definition and customer's characteristics in the future. This method is based on collective intelligence for finding a common consensus. How a business can gain knowledge through using DM techniques to support intelligent decision making in customer dynamics management is an important subject.

The RFM model is a common well-known method for customer dynamics segmentation. In [15,47,48], RFM can be considered as the most powerful and behavior-based model to implement CRM that extracts customer past information by using specific criteria. RFM values are defined as follows:

- R (Recency): How recently a customer has made a purchase. Customers who have recently purchased from you are more likely to purchase again from you.
- F (Frequency): How often a customer makes a purchase. The higher the frequency is the chances of customers responding to your offers, which indicates a higher degree of loyalty.
- M (Monetary): How much money a customer spends on purchases. Customers who have spent higher are more likely to purchase based on the offers, and the company should focus more on such customers.

RFM is a technique for customer behavioral analysis that can effectively investigate customer values. In [24], the RFM scoring model is used to transform the customer behavioral variables. By RFM scoring, customers are segmented into various target markets in terms of customer value. RFM valuables are important hidden variables in the database. In many studies, it has

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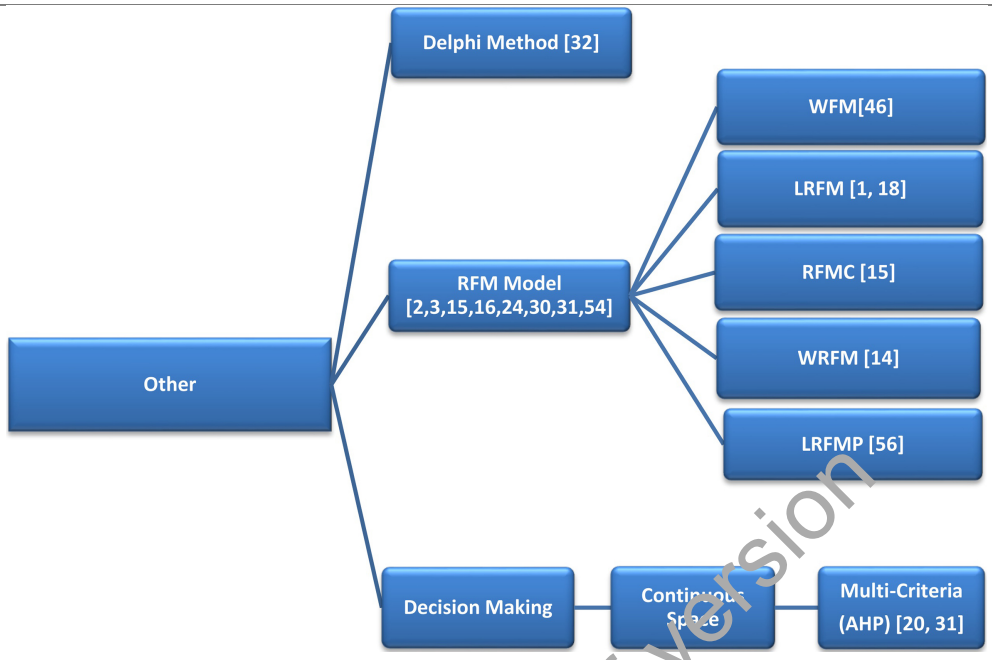


Fig. 6. Other techniques used in CRM, CLV and bank customer segmentation.

been shown that the higher R and F, the higher will be the probability of new transactions with the customer, and the higher M, the more will be the probability of customer return. Some researchers used this model for ranking customers of Export Development Bank in terms of credit. Therefore, in [31], RFM variables were extracted from the Export Development Bank's database and were accordingly normalized. Then, the variable weights were calculated using FAHP (Fuzzy AHP), and finally, a value for each customer was estimated. In [2], the RFM model is used to detect customer segment over 8 three-month periods. Moreover, in [3], the RFM values are used to classify three profitable groups of customers as follows: revolver user, transactor user, and convenience user.

LRFM is the extended version of RFM model that is used to take the length of the relationship into account. Authors in [18] represented a model to calculate customer life time value (CLV) based on LRFM customer relationship model, which consists of four dimensions: relation length (L), recent transaction time (R), buying frequency (F), and monetary (M) in banking services.

Some studies have added the count item to RFM model and implemented the RFMC model. The results revealed that the count item is not so useful and the outcome of RFM model was better than RFMC model [15]. Therefore, CLV is calculated based on weighted RFM method for each segment. In [56], an augmented RFM model called LRFMP to gain deeper and reasonable insight about customers are proposed.

A weighted RFM integrates AHP and DM into customer segmentation. Some researchers proposed WRFM (Weighted RFM) instead of RFM. They dedicated weights to R, F, and M depending on characteristics of the industry. These weights are different based on characteristics of the industry. For example, in some researches, placing the highest weighting on frequency, followed by recency has been suggested, with the lowest weighting on the monetary measure. The AHP method is a hierarchical process that is used to determine the weights of the RFM variables based on expert point of view in sales department. AHP is a well-known tool for decision-making in an analysis. In [20], researchers have proposed a combination of RFM with AHP and K-means algorithm and then segmented it with a group of customers from one of the big national banks in the country.

4.3. Evaluation of research's techniques

In the previous section, we studied DM technique used by some researchers for analyzing the customer behavior of banks and another finance industry. Now, we are going to evaluate these techniques based on the results of references. In Fig. 7, the usage percentage of different techniques in this literature is shown. As can

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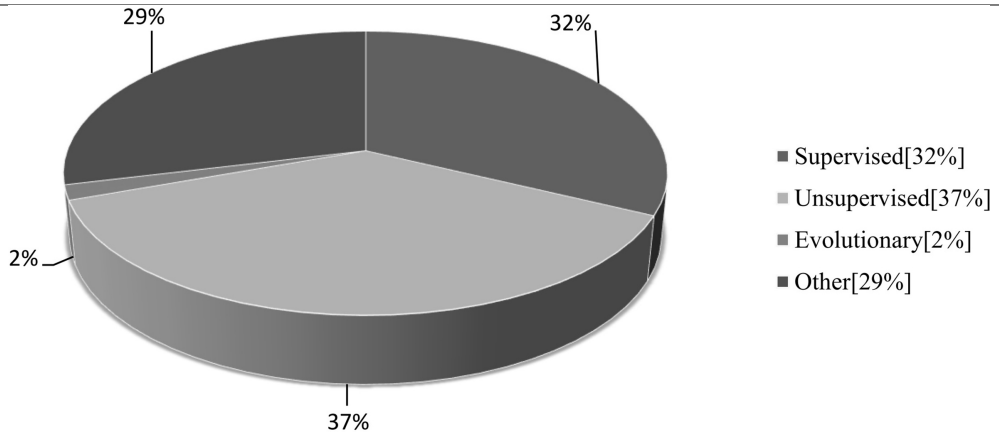


Fig. 7. Percentage of different DM techniques used in CRM, CLV and bank customer segmentation.

be seen in this Fig. 7, unsupervised learning techniques have the highest usage percentage in this literature.

4.3.1. Evaluation criteria

There are a number of criteria to evaluate DM techniques used in this study. We define these criteria and will discuss their obtained results. At first, we define *TP, TN, FP*, and *FN* according to the recommendation system [45].

- **True Positive (TP):** A product was recommended to a customer and the customer bought it.
- **False Positive (FP):** A product was recommended to a customer and the customer did not buy it.
- **False Negative (FN):** A product was not recommended to a customer and the customer bought it.
- **True Negative (TN):** A product was not recommended to a customer and the customer did not buy it.

Accuracy: This criterion is used to evaluate the performance of classification models. Classification accuracy is defined by Eq. (1) [45]:

$$Accuracy = \frac{TP + TN}{TP + TN + FN + FP} \tag{1}$$

Recall and Precision: These criteria are used to measure the quality in recommender and information retrieval systems and are defined as follows [45]:

$$Recall = \frac{TP}{TP + FN} \tag{2}$$

$$Precision = \frac{TP}{TP + FP} \tag{3}$$

Note that the increase in recommendation causes a decrease in *Precision* and an increase in *Recall*.

F-measure: This criterion leads to a balance in *Precision* and *Recall* [45]:

$$F\text{-}measure = (2 \times Recall \times Precision) / (Recall + Precision) \tag{4}$$

Support: Support is a measure of how frequently any given combination of antecedent and consequent occurs in a data set [45].

Confidence: It is defined by the percentage of cases, in which a consequent appears given that the antecedent has occurred. It essentially measures the strength of an association rule [45].

SSE: In some studies, sum of squared errors (SSE) [45] is investigated to calculate the appropriate number of clusters, which is shown in Eq. (5).

$$SSE = \sum_{i=1}^{k} \sum_{x \in c_i} dist^2 (m_i, x) \tag{5}$$

Where k is number of clusters, m_i is the cluster center, and x is the record belonging to the cluster.

If SSE is closer to 0, it indicates that the model has smaller random error component and is useful for prediction.

RMSE: This is a good measure of accuracy, which compares and defines the forecasting errors of different models by the following equation.

$$RMSE = \sqrt{MSE} \tag{6}$$

MSE: It is the mean square error or the residual mean square [45]:

$$MSE = \frac{SSE}{v} \tag{7}$$

For example, in [1], a combined algorithm (fuzzy c-means cluster and genetic algorithm) was proposed.

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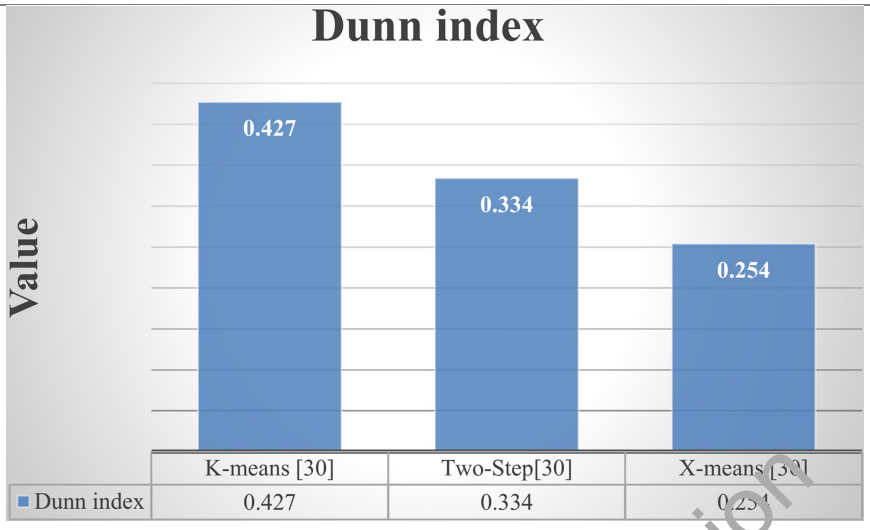


Fig. 8. Results of Dunn index for clustering techniques [30].

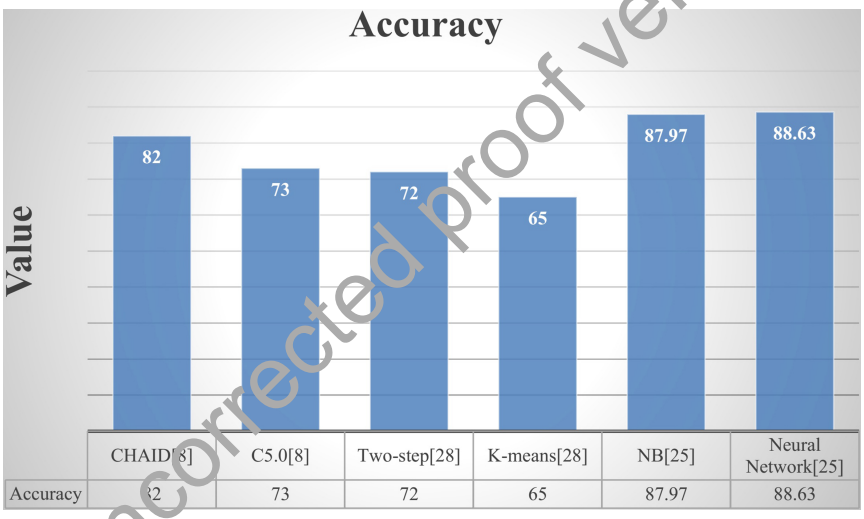


Fig. 9. Comparison the Accuracy of different techniques.

Table 2
Timeline of the researches in the literature

Algorithm	MSE
Fuzzy c-mean	0.0875
Combination of clustering and genetic algorithm	0.0152

Therefore, for comparing the efficiency of these algorithms, mean squared error (MSE) was used and its results are shown in Table 2. According to this study, it is clear that the combined clustering algorithm had a lower MSE; therefore, the authors suggested the use of this combination to obtain the most accurate cluster.

Dunn index: This metric is used for evaluating the clustering techniques. For a given assignment of clusters, a higher Dunn index indicates better clustering. For example, in [30], three techniques (including K-means, Two-Step, and X-means) are used for estimation of customer future value of different customer segments based on the adapted RFM model in retail banking context. In Fig. 8, results of this index are shown.

We studied the usage frequency of each criterion in the literature. Accordingly, the most frequently used criteria are *Accuracy, Precision*, *Recall*, and *F-measure*. As mentioned earlier, *Accuracy* is the most widely used criterion in all classes of DM techniques, because it provides a good insight for users. Figure 9 shows the accuracy results reported by the researches. This figure gives important information by comparing

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different techniques and approaches in terms of *Accuracy*.

According to the results presented in [28], if Two-step and K-means for customer segmentation are used to measure customer loyalty, the K-means algorithm has a poor function in recognizing the medium and very high loyalty levels. The two-step algorithm has a very low *Accuracy* for recognizing a low level of loyalty.

In [8,25], some DM techniques were used to classify customers and help the desired organization to make decisions. In [8], CAID and C5.0 decision tree techniques have been applied. According to Fig. 9, it is clear that the applied CHAID technique has more than 82% *Accuracy*, which is about 74% for C5.0 technique. As shown in Fig. 9, in [25], Neural Network and NB were used to predict the customer behavior and it was concluded that Neural Network is the best model for predictive performance with *Accuracy* rate of 88.63%.

5. Discussion and conclusion

According to our survey, numerous studies have been done on customer behavior until now. Some researchers believe that the importance of customer value in financial services industry is seldom realized. In banking industry, identifying the customers and their needs is an important matter. Assessment of the value of bank customers and determining their impact on the performance of banks are necessary to identify their key characteristics by using customer clustering. The banks can identify their most profitable customers and design marketing strategies for each group of customers by customer clustering. In this literature, we attempted to cover every aspect of this subject and almost reviewed all the studies conducted from 2001 to 2017, which are summarized in Appendix 2. In this paper, we conducted a research about data sets that have been used in recent studies. Moreover, we tried to review and summarize all DM techniques associated with analysis of bank customers. We reported experimental results presented by researches and the criteria used in them. Some studies have used a combination of DM techniques to increase *Accuracy* of their study. Moreover, there are many subjects in this domain, which needs further studies, and we are going to cite them in the following.

In today's dynamic environment, identifying customers' behavior is important for banking industry to gain a competitive advantage. The banks have to analyze their customers to identify their needs, which helps the banking industry to retain valuable customers and increase their CLV. Considering the importance of this subject, numerous studies have been done in this area in recent years. According to these studies, there are some issues that need more attention and we try to mention some of them in the following:

- Some researchers have used a limited data set; therefore, researchers should prepare a general data set to achieve a more complete analysis and make better decisions in future.
- Because *Accuracy* of customer clustering is an important matter, it is necessary that future studies use a combination of fuzzy clustering algorithms and evolutionary algorithms.
- Numerous studies have been conducted on CLV and the authors of these studies recommend researchers to work on more measures and comprehensively consider them or rather compare various CLV models in a specific industry.
- Some researchers used WRFM techniques and clustering algorithms based on customer's value to specify loyal and profitable customers; therefore, we recommend future researches to automatically set the weight of variables.

Appendix

Appendix 1: List of abbreviations

CRM	Customer relationship management
CLV	Customer lifetime value
DM	Data mining
SOM	Self-Organizing map
AHP	Analytic hierarchy process
CART	Classification and regression trees
CHAID	Chi square automatic interaction detection
KNN	K Nearest neighbor
CF	Clustering features
NB	Naïve Bayes
MLPNN	Multilayer perception neural network
BIC	Bayesian information criterion
AIC	Akaike information criterion
GSP	Generalized sequential pattern
WRFM	Weighted RFM
ARM	Association Rule Mining
RS_IP	Recommendation system using CPM_IP
RS_IB	Recommendation system using CPM_IB
PS_IGB	Recommendation system using CPM_IGB

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Appendix 2: Summery of the researches used in the literature

Reference	Category	Used technique	Goal	Result
[1]	Unsupervised/ Evolutionary	Genetic Algorithms, Fuzzy C-Means Clustering	Identifying customers and customers' needs is an important factor, which help service providers and producers to gain competitive advantage in providing their products/services to the customers	Two clusters of customers are reviewed. In the first cluster, customers with a higher duration of the relationship, recency of trade, and frequency of trade but lower monetary value can be considered as loyal customers and certain customers. In the second cluster, customers with a higher recency of trade and monetary value but lower length of the relationship and frequency of trade can be considered as newcomers and uncertain customers. Moreover, it is revealed that combined clustering algorithm has a lower MSE and a higher run time compared to the Fuzzy C-means algorithm.
[22]	Supervised/ Unsupervised	CF tree, K-nearest neighbors	Presenting a new method to increase the accuracy and quality of recommendations aimed at attracting more customers	The results indicated that this proposed combination of CF method with K-nearest neighbors with higher values for each cluster has a higher accuracy.
[8]	Supervised	CHAID, C5.0	Keeping the cost in a lower level and increasing the revenue for the firm	The extracted rules illustrated the improvements of marketing strategy that should be done to encourage and attract them to purchase optional services.
[60]	Supervised	CART	Providing CLV model to carefully assess and classify banking customers using individual measures and covering customers' relationships with a portfolio of products of the company	The results indicated that the model implements a value-based segmentation, which helps banks to facilitate the process of customer management.
[2]	Unsupervised/ Other	GSP algorithm, RFM model, K-means algorithm	Providing DM techniques to support intelligent decision making in customer dynamics management	To investigate the impact of customer dynamics on segments' structural changes, a method was successfully implemented on the customer's data of a private bank.
[27]	Supervised/ Unsupervised	Chain Modeling, CART	Providing a model-based on a combination of first-order Markov chain modeling and CART for customer valuation in cooperation with a leading German retail bank	A retail bank could use this model to assign acquisition allowances for new customers by comparing prospective with existing customers and estimating their lifetime value.
[29]	Supervised	Neural Network	Proposing a method to manage the existing customers by using misclassification patterns of credit scoring model	It is attempted to propose management strategies appropriate for the group characteristics because credit and behavioral scoring have become popular tools to predict the financial risk of loan customers and to help loan companies to deal with the customers.
[3]	Supervised/ Other	Neural Network, RFM	Providing a general integrated DM and behavioral scoring model for customer behavior analysis	This two-stage behavioral scoring model is used for analyzing bank databases; for example, by understanding customer value, the bank gains the opportunity to establish better customer relationships while increasing customer loyalty and revenue, which can be applied to predict personal bankruptcy among bank customers to the account database.
[24]	Other	RFM scoring approach	Proposing an approach for mining changes in customer behavior to assist managers in developing better marketing strategies and better understand the changes in customer needs	Mining changes for customer behavior are useful for satisfying customer needs in dynamic business environments. Mining change can extract further value from customer, product, and transaction databases.

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Reference	Category	Used technique	Goal	Result
[10]	Supervised	Decision Tree	Proposing a framework for analyzing customer value and segmenting customers based on their value	Three perspectives on customer value such as current value, potential value, and customer loyalty assist marketing managers in identifying customer segmentation with more balanced viewpoints.
[53]	Supervised	MLPNN, NB	Proposing an efficient CRM-data mining framework used to predict the behavior of customers	The results indicated that, in comparison with NB, MLPNN is the best model that achieves high predictive performance and high accuracy rate.
[57]	Supervised	C4.5, NB	Proposing a method, which uses model-based approach over partitioning to predict customer's decision and improve the return rate for bank direct marketing	The results indicated that partitioning based method increases accuracy rate, predicted true positive and true negative customer number. In addition, the accuracy of C4.5 is better than Naive Bayes method.
[4]	Supervised/ Unsupervised	C5.0, SOM algorithm	Understanding the relationships between churns is essential in creating effective retention policy models for dealing with 'churners'	The construction process signified an interesting and important approach toward a better support in retaining possible churners.
[32]	Other	Delphi method	Providing an overview of customer relationship management that is strongly influenced by corporate culture, corporate identity, and employees	New characteristic of CRM was profiled as philosophy and business strategy. CRM should be created on both sides: company's side and customer's side as well.
[26]	Supervised	Neural network, CHAID	Using the DM techniques to be able to identify valuable customers, predict future behaviors, and enable firms to make proactive, knowledge-driven decisions	With full implementation of a CRM program, organizations faster improved loyalty, increase the value of their customers, and attract the right customers.
[15]	Unsupervised/ Other	RFMC, K-means clustering	Building successful customer relationship management can provide basic information to deploy more targeted and personalized marketing	The results indicated that adding the count item as a new parameter to RFM method makes no difference with clustering results, so CLV is calculated based on weighted RFM method for each segment.
[30]	Unsupervised/ Other	RFM, K-means clustering	Providing a framework for estimating future value of each segment with analyzing the trend of customer value in different seasons by time series method	CLV can calculate the past and present value of the customers and predict the future value of customers.
[14]	Unsupervised/ Other	WRFM, K-means clustering	Clustering the customers based on CLV and then recommending the product to different groups of customers using association rule mining technique	The results indicated that the rule support and confidence for loyal and new customers is relatively high, so this method is useful and reliable for these groups. But CF method may be applicable for the disloyal segment.
[31]	Unsupervised/ Other	RFM, FAHP, K-means clustering	Using credit and behavioral models with K-means algorithm for ranking the bank customers in terms of credit	The results indicated that this method can help banks, as well as finance and credit institutions to purposefully rank their customers.
[21]	Unsupervised	K-means clustering, Apriori	Providing a new two-stage framework of customer behavior	This method helps the bank to simply understand customer value; the bank gains the opportunity to establish better customer relationships.
[16]	Unsupervised/ Other	K-means clustering, RFM	Using K-means clustering approach to determine customer's CLV and segment them based on RFM measures	This study can suggest a CLV model considering the recency, frequency and monetary at the same time. It clustered customers into segments according to their lifetime value expressed in terms of RFM.

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Reference	Category	Used technique	Goal	Result
[56]	Unsupervised/ Other	K-means clustering, LRFMP	Proposing a new model called LRFMP (Length, Recency, Frequency, Monetary and Periodicity) for classifying customers	The results indicated that LRFMP provides valuable insights about behaviors of different customer types in the Turkish grocery industry.
[20]	Unsupervised/ Other	AHP, K-means clustering	Providing a framework that helps the bank to recognition and retention of valuable customers is great importance for the banks and other financial institutions	The results indicated that this proposed method can be utilized as an applicable, effective, and accurate method in identifying the customers of banks to identify pioneer profitable customers.
[33]	Other	RFM	Proposing a tree structure (called an RFM-pattern-tree) to compress and store entire transactional database, as well as developing a pattern growth-based algorithm (known as RFMP-growth) to discover all the RFM-patterns in an RFM-pattern-tree	The results indicated that RFMP-growth can efficiently discover more valuable patterns than conventional frequently used pattern mining algorithms.
[54]	Other	RFM	Presenting a comparison between the traditional methods for the calculation of customer relationship performance and the modern metrics like customer-centered	The modern metrics were judged to be more effective and useful and profitable in the shipping sector of Greek industrial activity
[18]	Unsupervised/ Other	LRFM, K-means clustering	Introducing a model to calculate CLV based on LRFM customer relationship model	This study can help assist managers of bank branches to identify profitable customers and prioritize them. Thus, this study is considered useful for customer relationship management.
[28]	Supervised/ Unsupervised/ Other	RFM, K-means clustering, C5 model	Proposing RFM technique and clustering algorithms for customer segmentation based on customer's value to specify loyal and profitable customers	After analysis and evaluation of rules obtained from C5, it is concluded that applying the C5 model on the clusters of K-means algorithm has a poor function in recognizing medium and very high loyalty levels.
[7]	Unsupervised/ Other	Fuzzy c-means, AHP, WRFM	Proposing a new method of customer segmentation and ranking by combining fuzzy clustering (as a segmentation method) and fuzzy AHP (as a ranking method)	Companies can implement customer-centric strategies for maximizing each group's lifetime profit, as well as customer loyalty. Managers can consider this proposed CLV calculation methodology for selling the next best services/products to the group of customers.
[46]	Unsupervised/ Other	ARM, WFM	Proposing a new algorithm for discovering frequent item sets with high revenues from FM-weighted transactions with customer analysis	The results indicated that this proposed approach can enable the discovery of interesting and valuable patterns that have not been discovered using conventional approaches. In addition, the top k frequent item sets discovered using this approach outperformed those discovered using the conventional approach in the prediction of revenues from customers in next-period transactions.

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